

COMPUTER SCIENCE & IT

DIGITAL LOGIC



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BOOLEAN THEOREMS AND GATES -1

Lecture : 02



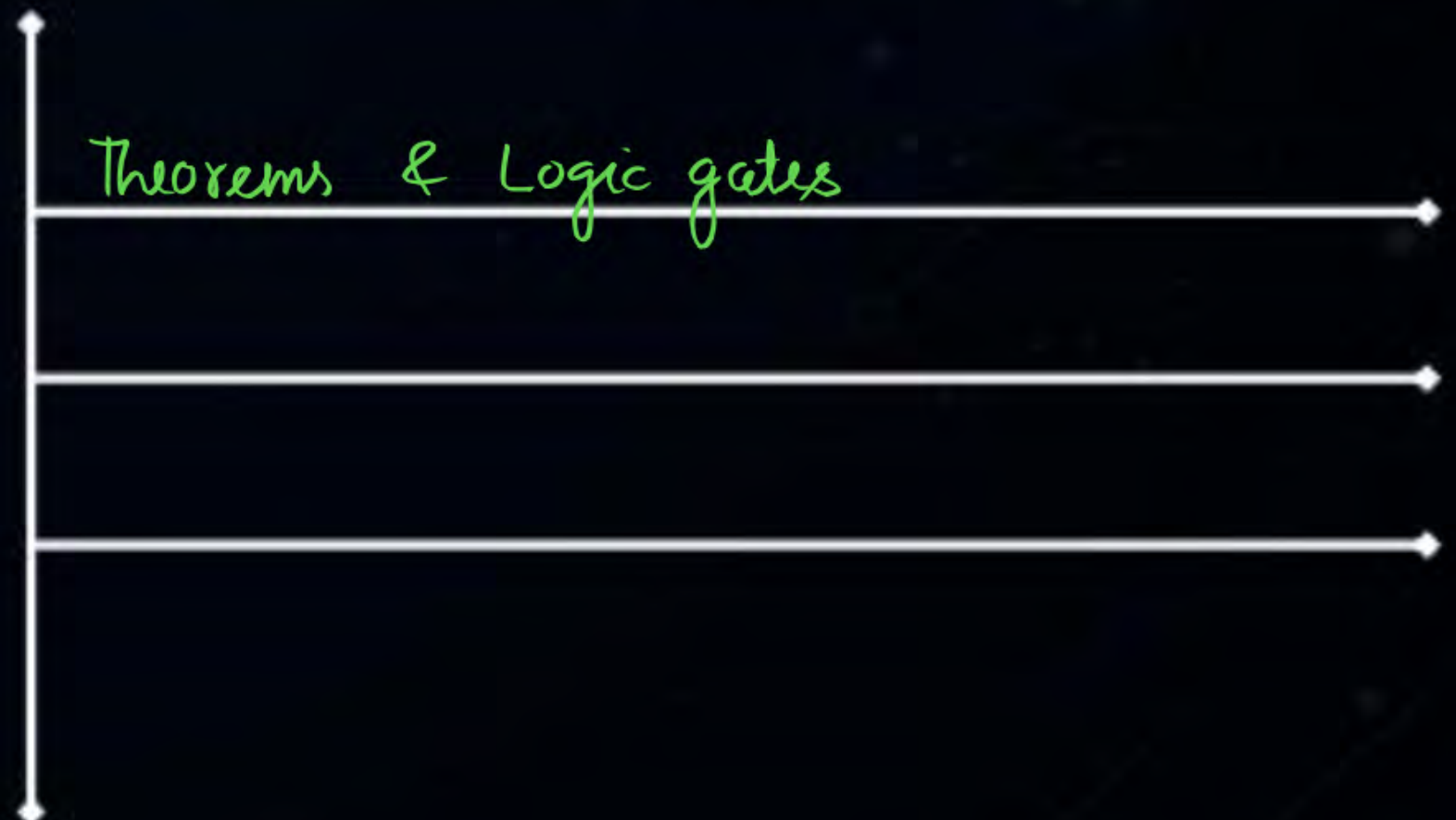
Recap of Previous Lecture

Imp basics & Theorems:



Topics to be Covered

Theorems & Logic gates



3. Very imp Boolean theorems

- Redundancy Theorem:

- $$AB + \bar{A}C + BC = AB + \bar{A}C$$



 redundant term

- $$AB + \bar{A}\bar{C} + \bar{B}\bar{C} = AB + \bar{B}\bar{C}$$



 redundant

- $$\bar{A}B + \bar{A}C + B\bar{C} = \bar{A}C + B\bar{C}$$



 redundant term

- $$AB + \bar{B}C + \bar{A}C \Rightarrow N.A.$$

- $$A\bar{B}C + AB + AC = A\bar{B}C + AB$$



 redundant term

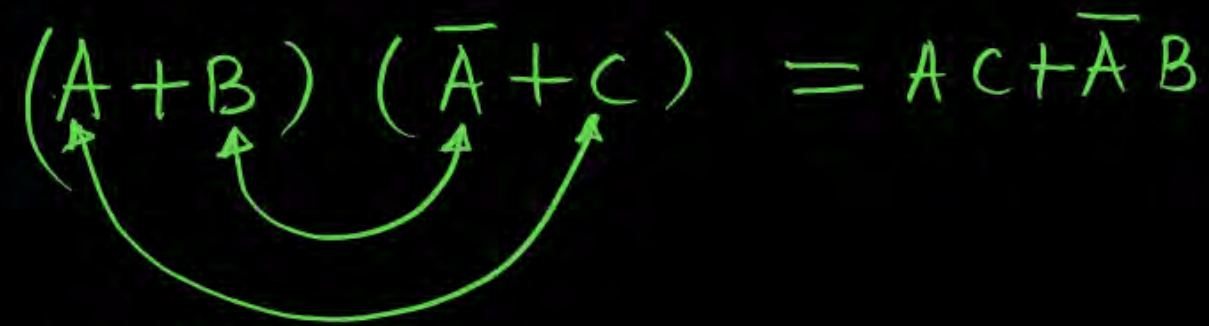
$$= A\bar{B}C + AB$$

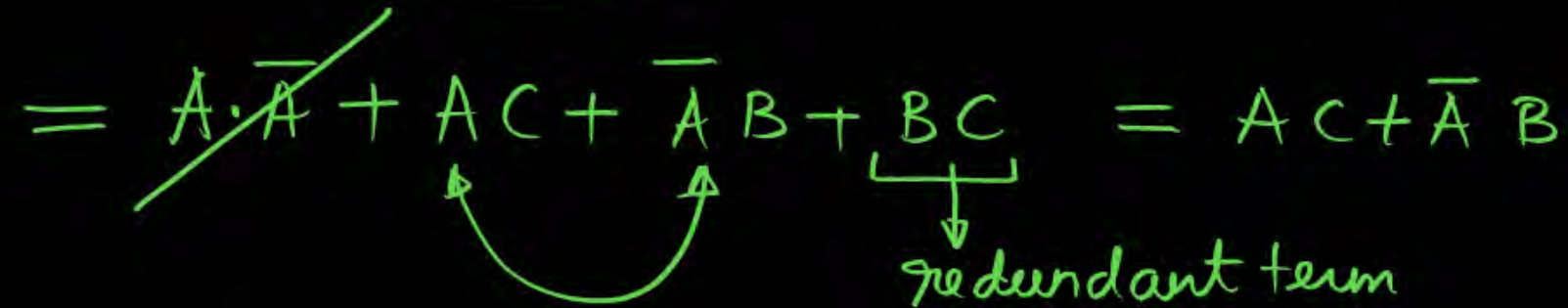
$$= A(\bar{B}C + B)$$

$$= A(B + C)$$

$$= AB + AC$$

- Transpose Theorem:

$$(A+B)(\bar{A}+C) = AC + \bar{A}B$$


$$= \cancel{A \cdot \bar{A}} + AC + \bar{A}B + \underbrace{BC}_{\text{redundant term}} = AC + \bar{A}B$$


$$\rightarrow \underline{AC} + \underline{\bar{A}B} = (A+B)(\bar{A}+C)$$

$$\rightarrow \bar{A}B + A\bar{C} = (\bar{A}+\bar{C})(A+B)$$

$$\rightarrow \bar{A}\bar{B} + B\bar{C} = (\bar{B}+\bar{C})(B+\bar{A})$$

$$\bullet (A + \overline{B} + C) (A + \overline{B} + \overline{C}) (\overline{A} + B + \overline{C})$$

$$= \left[(\underline{A} + \underline{\overline{B}}) + 0 \right] \left[\underline{\overline{A}} + B + \overline{C} \right]$$

$$= A \cdot (B + \overline{C}) + \overline{A} \cdot \overline{B}$$

$$= AB + A\overline{C} + \overline{A}\overline{B} \quad \checkmark$$

$$\bullet (A + BC) (A + B + \overline{C}) (A + \overline{B} + \overline{C})$$

$$(A + BC) (A + \overline{C})$$

$$A + BC \cdot \overline{C}$$

$$= A \quad \checkmark$$

[DeMorgan Theorem]



- $\overline{A+B+C} = \bar{A} \cdot \bar{B} \cdot \bar{C}$

- $\overline{A \cdot B \cdot C} = \bar{A} + \bar{B} + \bar{C}$

- $\overline{AB+BC} = \overline{A \cdot B} \cdot \overline{B \cdot C} = (\bar{A} + \bar{B}) \cdot (\bar{B} + \bar{C})$

- $$\begin{aligned} \overline{(A+\bar{C}) \cdot (B+\bar{A})} &= \overline{A+\bar{C}} + \overline{B+\bar{A}} \\ &= \bar{A} \cdot C + \bar{B} \cdot A \end{aligned}$$

[Question]



$A + BC + \bar{A}C$ is equal to

☒ (a) $(A + C)$

(b) $(A + B)(B + C)$

(c) $(A + B)(\bar{A} + C)$

(d) $C(\bar{A} + B)$

$$A + BC + \bar{A}C$$

$$= A + \bar{A}C + BC$$

$$= A + C + BC$$

$$= A + C(1 + B)$$

$$= A + C \cdot 1$$

$$= A + C$$

[Question]



$\bar{A}B$ + AC + $\bar{B}C$ is equivalent to

(a) $\bar{A}B + AC$ ✗

(b) ✓ $\bar{A}B + C$

(c) $AC + \bar{B}C$ ✗

(d) $\bar{A}B + \bar{B}C$ ✗

$$\bar{A}B + AC + \bar{B}C$$

$$= \bar{A} \cdot B + (A + \bar{B}) C$$

$$= P + \bar{P} C$$

$$= (P + \bar{P}) \cdot (P + C)$$

$$= P + C$$

$$= \bar{A}B + C = (\bar{A} + C) \cdot (B + C)$$

$$P = \bar{A} B$$

$$\bar{P} = \overline{\bar{A} \cdot B} = (A + \bar{B})$$

[Question]



$A\bar{B}$ ($\bar{A}B + \bar{B}C + \underline{A\bar{B}D} + \underline{A\bar{B}\bar{D}}$) is equal to

(a) ✓✓ $(\bar{A} + B)$

(b) $(\bar{A} + B)(B + \bar{C})$

(c) $(\bar{A} + B)(A + \bar{B})$

(d) $\bar{A}B + \bar{B}D$

$A\bar{B}$

$A\bar{B} [A\bar{B} + \text{anything}]$

$= \overline{A\bar{B}}$
 $= (\bar{A} + B)$

$P \cdot (P + \text{anything}) = P$

$\rightarrow P \cdot P + P \cdot \text{anything}$

$= P + P \cdot \text{anything}$

$= P[1 + \text{anything}]$

$= P \cdot 1 = P$

$$A\bar{B}C [\bar{A}\bar{B}\bar{C} + \bar{B}\bar{C}\bar{D} + BCD + \bar{B}C]$$

$$A \cdot \bar{B}C \cdot [\bar{B}C + \text{anything}]$$

$$= A \cdot \bar{B}C$$

[Question]

$$\overline{(\bar{A} + \bar{B})(\bar{B} + \bar{C})} \text{ is equal to } \begin{aligned} &= \overline{\bar{A} + \bar{B}} + \overline{\bar{B} + \bar{C}} \\ &= AB + BC \\ &= B[A + C] \end{aligned}$$

(a) $\bar{B}(A + C)$

(b) $A(B + C)$

(c) ☒ $B(A + C)$

(d) $C(A + B)$

[Question]



$\bar{A}\bar{B} + AC + \bar{B}C$ is equivalent to $= \bar{A}\bar{B} + AC$
Handwritten notes: A green arrow points from $\bar{A}\bar{B}$ to AC , and another green arrow points from $\bar{B}C$ to the word "redundant".

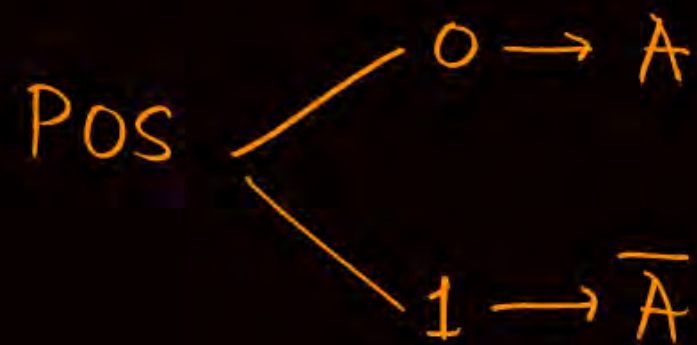
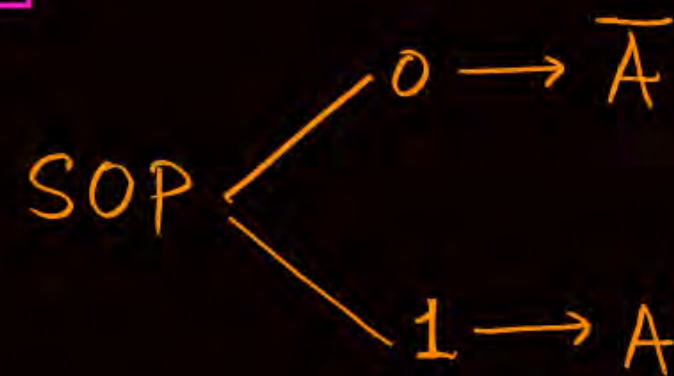
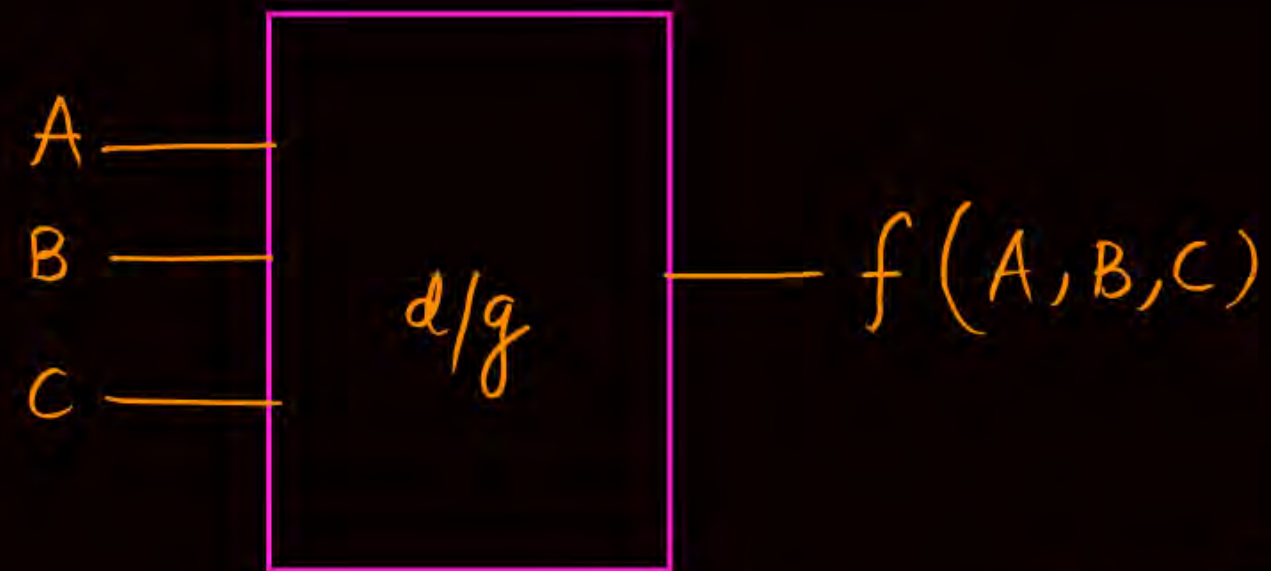
(a) $(A + \bar{B}) \cdot (\bar{A}\bar{B} + C)$

☒ (b) $\bar{A}\bar{B} + AC$

(c) $AC + \bar{B}C$

(d) $\bar{A}\bar{B} + \bar{B}C$

	A	B	SOP	POS	$f_1(A, B)$
0	0	0	$\overline{A} \cdot \overline{B}$	$(A + B)$	1
1	0	1	$\overline{A} \cdot B$	$(A + \overline{B})$	0
2	1	0	$A \cdot \overline{B}$	$(\overline{A} + B)$	1
3	1	1	$A \cdot B$	$(\overline{A} + \overline{B})$	0



$$\begin{aligned}
 f_1(A, B) &= \sum (0, 2) = \overline{A} \cdot \overline{B} + A \cdot \overline{B} = \overline{B} \\
 &= \prod (1, 3) = (A + \overline{B}) \cdot (\overline{A} + \overline{B}) \\
 f_1(0, 0) &= 1 & f_1(1, 0) &= \overline{B} + A \cdot \overline{A} = \overline{B} \\
 f_1(0, 1) &= 0 & & \\
 &= 1 & & \\
 f_1(1, 1) &= 0 & &
 \end{aligned}$$

$$\begin{aligned}
 \bullet f(A, B, C) &= \Sigma(0, 1, 4, 5, 6) = \Pi(2, 3, 7) = (A + \bar{B} + C) \cdot (A + \bar{B} + \bar{C}) \\
 &\quad (\bar{A} + \bar{B} + \bar{C}) \\
 &= \underbrace{\bar{A} \bar{B} \bar{C}}_{000} + \bar{A} \bar{B} C + A \bar{B} \bar{C} \\
 &\quad + A \bar{B} C + AB \bar{C}
 \end{aligned}$$

$$\bullet \underline{f(A, B, C) = \Sigma(0, 5, 6) = \Pi(1, 2, 3, 4, 7)}$$

$$\underline{f(A, B, C)} = \overline{f(A, B, C)} = f_1(A, B, C) = \Sigma(1, 2, 3, 4, 7) = \Pi(0, 5, 6)$$

$$f(A, B, C) = A\bar{B} + \bar{B}C + ABC$$

$$= A\bar{B}(\bar{C} + C) + (\bar{A} + A)\bar{B}C + ABC$$

$$= A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C + A\bar{B}C + ABC$$

$$\begin{matrix} 100 & 101 & 001 & 101 & 111 \end{matrix}$$

$$= \Sigma(1, 4, 5, 7)$$

$$= \Sigma(1, 4, 5, 7)$$

$A\bar{B}$

$$100 \rightarrow 4$$

$$101 \rightarrow 5$$

$\bar{B}C$

$$001 \rightarrow 1$$

$$101 \rightarrow 5$$

$$f(A, B, C, D) = \overline{B}\overline{D} + \overline{A}C + A\overline{B}D$$

	$\overline{B}$	$\overline{D}$	
0	0	0	$\rightarrow 0$
0	0	1	$\rightarrow 2$
1	0	0	$\rightarrow 8$
1	0	1	$\rightarrow 10$

$\overline{A}$	C	
0	0	$\rightarrow 2$
0	1	$\rightarrow 3$
1	1	$\rightarrow 6$
1	1	$\rightarrow 7$

A	$\overline{B}$	D	
1	0	0	$\rightarrow 9$
1	0	1	$\rightarrow 11$

$$= \sum(0, 2, 3, 6, 7, 8, 9, 10, 11)$$

$$f(A, B, C, D) = ABC + \overline{B}C + \overline{A}\overline{D}$$

$$= \sum (0, 2, 3, 4, 6, 10, 11, 14, 15)$$

A B C

$$1 \ 1 \ 1 \ 0 \rightarrow 14$$

$$1 \ 1 \ 1 \ 1 \rightarrow 15$$

B C

$$0 \ 0 \ 1 \ 0 \rightarrow 2$$

$$0 \ 0 \ 1 \ 1 \rightarrow 3$$

$$1 \ 0 \ 1 \ 0 \rightarrow 10$$

$$1 \ 0 \ 1 \ 1 \rightarrow 11$$

$\overline{A}$

$\overline{D}$

$$0 \ 0 \ 0 \ 0 \rightarrow 0$$

$$0 \ 0 \ 1 \ 0 \rightarrow 2$$

$$0 \ 1 \ 0 \ 0 \rightarrow 4$$

$$0 \ 1 \ 1 \ 0 \rightarrow 6$$

$$\bullet f_1(A, B, C) = (\overline{A} + B) (B + \overline{C}) (A + C)$$

$\overline{A} + B$

$B + \overline{C}$

$A + \quad + C$

$$1 \ 0 \ 0 \rightarrow 4$$

$$0 \ 0 \ 1 \rightarrow 1$$

$$0 \ 0 \ 0 \rightarrow 0$$

$$1 \ 0 \ 1 \rightarrow 5$$

$$1 \ 0 \ 1 \rightarrow 5$$

$$0 \ 1 \ 0 \rightarrow 2$$

$$= \prod (0, 1, 2, 4, 5) = \sum (3, 6, 7)$$

H.W. SOP/POS of following functions.

- $f(A, B, C, D) = \overline{A}B + \overline{B}CD + AD + BD$

- $f(A, B, C) = \overline{(A+B)(\overline{B}+C)} + AC$

- $f(A, B, C, D) = \overline{(A+B)(\overline{C}+\overline{D})} + AB\overline{C} + BD$

- If $f(A, B, C) = \overline{A}\overline{B} + BC$

then $f_1(A, B, C) = \overline{f}(A, B, C) = \sum ? = \pi(?)$



2 Minute Summary

→ Theorems & grp basics.

Thank you

GW
Soldiers !

